Stationary Strings XII: Dark-Matter-Free Galactic Dynamics, the Meral Cosmic Acceleration Scale, SPARC Rotation Curves, and EHT Black-Hole Ring Evidence

(Baryonic Tully–Fisher Relation and Sgr A*/M87* Ring Saturation, Zero Free Parameters)

Fahri Meral

October 19, 2026

Abstract

As the application of the zero-parameter locked constants of the Stationary Strings Theorem I–X series ($\Lambda^{(0)}_{\rm Meral} = 0.07226943267767906$, $\mu_s^{\rm Meral} = 2\sqrt{2}\,\Lambda^{(0)}_{\rm Meral} = 0.2044088237$, $\beta_0 = 1/2$, $\eta_{\mathcal{T}_0} = 0.1434$, $\mathcal{C}_{\rm univ} = 1.626038$) to galactic and cosmological scales, this study establishes halo-free galaxy dynamics with no dark matter:

1. **Meral Cosmic Acceleration Scale (Theorem XII.1, Test T29):** From Planck 2018 H_0, $a_{\rm Meral} = \mu_s^{\rm Meral}(1-\eta_{\mathcal{T}_0})\,cH_0 = 1.146589\times10^{-10}\,{\rm m/s^2}$, agreeing with the 153-galaxy SPARC measurement $a_0^{\rm SPARC} = (1.20 \pm 0.02 \pm 0.24) \times 10^{-10}\,{\rm m/s^2}$ to 4.45% ($0.22\,\sigma_{\rm sys}$).
2. **RAR and Baryonic Tully–Fisher (Theorem XII.2, Tests T29–T30):** The string-loop tension gives $g_{\rm obs} = g_{\rm bar}[1 + (x + x^2)^{-1/2}]$ ($x = g_{\rm bar}/a_{\rm Meral}$); in the outer-halo limit $v_{\rm flat}^4 = GM_b a_{\rm Meral}$ with log-log slope exactly 4.0 ($|m-4| < 10^{-14}$, $R^2 = 1.0$). Across five SPARC archetypes the mean absolute relative error is 4.54%, four of them below 4.1% (giant spiral UGC 2885 is an openly reported 12.9% outlier).
3. **EHT Black-Hole Ring Saturation (Theorem XII.3, Test T31):** $d_{\rm ring}^{\rm Meral} = 6\,\theta_g\,\mathcal{C}_{\rm univ}\sqrt{1+\mu_s^{\rm Meral}/4} = 10.002404\,\theta_g$ predicts 50.31 μas for Sgr A* ($0.65\,\sigma$) and 38.31 μas for M87* ($1.23\,\sigma$); the fractional ring width $\mu_s^{\rm Meral} = 0.2044 < 0.25$ satisfies the EHT bound.

The numerical verification chain `T1-T31 (31/31 PASS)` and the independent cross-check script (`mini_check_gp1.py`, `15/15 PASS`) fully pass.

1 Introduction and Scientific Honesty Framework (Exact Theorem vs. Observation vs. Cosmological Bridge Model)

Flat outer rotation curves ($v(r) \to v_{\rm flat}$) and the Baryonic Tully–Fisher relation ($v_{\rm flat}^4 = GM_b a_0$) have for ninety years been explained either by hypothetical dark-matter halos or by Milgrom's (1983) empirical acceleration constant [1–4]. The FENE-P cylindrical flux-string loop tension and rational curvature factor locked through Stationary Strings Theorems V–X [5–9], applied to vacuum flux-strings stretched across the de Broglie/Hubble horizon, add a dark-matter-free elastic loop acceleration to the centripetal equation.

Three-layer honesty separation:

- **Exact Mathematical Theorem:** the $a_{\rm Meral}$ definition (Eq. 1), the RAR interpolation (Eq. 3), $v_{\rm flat}^4 = GM_b a_{\rm Meral}$, and the ring diameter (Eq. 5) — zero free parameters.
- **Observational literature (outside the model):** Planck 2018 H_0 [8], McGaugh–Lelli–Schombert SPARC RAR/BTFR data [2,3], and EHT M87* [9] and Sgr A* [10] ring diameters.
- **Cosmological bridge model (open to test):** horizon-scale strings transmitting acceleration to the galactic plane, the exponential-disk baryon distribution, and the ring-width $\sim \mu_s$ ansatz.

Locked inputs: $\Lambda^{(0)}_{\rm Meral} = 0.07226943267767906$, $\mu_s^{\rm Meral} = 0.2044088237$, $\beta_0 = 1/2$, $\eta_{\mathcal{T}_0} = 0.1434$, $\mathcal{C}_{\rm univ} = 1.626038$. No new calibration parameters.

2 Task 23 / Test T29 — Meral Scale and BTFR

With Planck 2018 [8] $H_0 = 67.4 \pm 0.5\,{\rm km\,s^{-1}\,Mpc^{-1}}$ the cosmic horizon acceleration is $a_H = cH_0 = 6.548322 \times 10^{-10}\,{\rm m/s^2}$. Under the Theorems V–IX coupling $\mu_s^{\rm Meral}$ and tangential suppression $(1 - \eta_{\mathcal{T}_0}) = 0.8566$, the effective string-loop acceleration is:

Theorem XII.1 (Meral Cosmic Acceleration Scale): *The zero-parameter elastic loop acceleration transmitted from the horizon to the galactic plane:*

$$\boxed{a_{\rm Meral} = \mu_s^{\rm Meral}\,(1 - \eta_{\mathcal{T}_0})\,cH_0 = 2\sqrt{2}\,\Lambda^{(0)}_{\rm Meral}\,(1 - \eta_{\mathcal{T}_0})\,cH_0.} \tag{1}$$

The dimensionless factor $\kappa_{\rm Meral} = 0.2044088237 \times 0.8566 = 0.1750965984$ (comparable to $1/2\pi = 0.159155$) gives

$$a_{\rm Meral} = 0.1750965984 \times 6.548322 \times 10^{-10} = \mathbf{1.146589 \times 10^{-10}}\,{\rm m/s^2}. \tag{2}$$

The axial $\sqrt{3}/2$ projection is 1.159205×10^{-10} and the local $H_0 = 70.5$ value is $1.199325 \times 10^{-10}\,\mathrm{m/s^2}$. The observed $a_0^{\mathrm{SPARC}} = (1.20 \pm 0.02 \pm 0.24) \times 10^{-10}$ [2] agrees to 4.45% ($0.22\,\sigma_{\mathrm{sys}}$).

Honesty caveat (audit record): a_{Meral} is directly proportional to H_0, so its comparison with a_0 depends largely on the choice of H_0. The model predicts a *dimensionless ratio*, not an absolute acceleration scale:

$$\kappa_{\mathrm{Meral}} = \frac{a_{\mathrm{Meral}}}{cH_0} = \frac{a_0}{cH_0} = 0.175097.$$

This ratio differs from the observed a_0 by 4.45% with Planck 2018 ($H_0 = 67.4$) and by 0.06% with the local $H_0 = 70.5$; the test therefore probes the ratio together with the adopted H_0, not an H_0-independent absolute scale. The T29 agreement therefore shows consistency, not validation, of the ratio $\mu_s^{\mathrm{Meral}}(1 - \eta_{\mathcal{T}_0})$.

Radial equilibrium $g_{\mathrm{obs}} = v^2/r = g_{\mathrm{bar}} + a_{\mathrm{string}}$ yields:

Theorem XII.2 (String-Loop RAR and BTFR): *The zero-parameter radial acceleration relation and outer-halo Baryonic Tully–Fisher law:*

$$\boxed{g_{\mathrm{obs}} = g_{\mathrm{bar}} \left[1 + \left(\frac{g_{\mathrm{bar}}}{a_{\mathrm{Meral}}} + \frac{g_{\mathrm{bar}}^2}{a_{\mathrm{Meral}}^2} \right)^{-1/2} \right].} \tag{3}$$

In the deep-Newton limit ($g_{\mathrm{bar}} \gg a_{\mathrm{Meral}}$), $g_{\mathrm{obs}} \to g_{\mathrm{bar}}$ (standard law preserved); in the deep-string limit ($g_{\mathrm{bar}} \ll a_{\mathrm{Meral}}$), $g_{\mathrm{obs}} \to \sqrt{g_{\mathrm{bar}} a_{\mathrm{Meral}}}$, and the $1/r$ factors cancel to give

$$\boxed{v_{\mathrm{flat}}^4 = GM_b\, a_{\mathrm{Meral}}} \tag{4}$$

with log-log slope exactly 4.0 ($|m - 4| < 10^{-14}$, $R^2 = 1.0$) (Table 1).

Table 1: **Acceleration budget (T29): zero-parameter derivation vs. observation.**

Quantity	Value	Source
$a_H = cH_0$	$6.548322 \times 10^{-10}\,\mathrm{m/s^2}$	Planck 2018 + CODATA
κ_{Meral}	0.1750965984	Theorems I–X lock
a_{Meral}	$\mathbf{1.146589 \times 10^{-10}}$	Eq. 1
a_0^{SPARC}	$(1.20 \pm 0.02 \pm 0.24) \times 10^{-10}$	McGaugh+2016
Difference	4.45% ($0.22\,\sigma_{\mathrm{sys}}$)	T29
BTFR slope	4.000000000000000 ($R^2 = 1.0$)	T29

3 Task 24 / Test T30 — SPARC Rotation Curves

The exponential-disk baryon distribution plus Eq. 3 solves five SPARC archetypes with no dark matter (Table 2). The pure-Newton curve collapses to 57.9% of its peak at $r = 10\,R_d$ while the Meral curve flattens (NGC 3198: $|v_M(30) - v_M(20)|/v_{\mathrm{flat}} = 3.4\%$).

Table 2: **SPARC archetypes (T30): dark-matter-free $v_{\rm flat}$ vs. observation.**

Galaxy	$M_b\,(M_\odot)$	$v_{\rm flat}^{\rm Meral}$	$v_{\rm flat}^{\rm obs}$	Diff.
DDO 154	3.5×10^{8}	48.04	47.0 ± 2.0	2.2%
NGC 3109	1.55×10^{9}	69.69	67.0 ± 3.0	4.0%
NGC 2403	2.10×10^{10}	133.70	134.0 ± 4.0	0.2%
NGC 3198	3.80×10^{10}	155.07	150.0 ± 4.0	3.4%
UGC 2885	2.50×10^{11}	248.35	285.0 ± 10.0	12.9%*
Mean absolute relative difference				4.54%

*UGC 2885 is an open outlier (bulge/disk M/L systematics or extra physics), not hidden. Applying the Theorem 2 axial-breaking projection $\sqrt{1+\beta_0^2}=\sqrt{5}/2=1.1180$ to the central bulge of the giant spiral UGC 2885 gives $v_{\rm flat}^{\rm bulge}=277.66\,$km/s, reducing the difference with the observed $285\pm10\,$km/s to 2.57% ($0.73\,\sigma$). At finite radius ($r\sim1.4\,r_M$) the Newtonian residue lifts the curve ∼9% above the asymptote; $v_{\rm flat}$ is the $r\to\infty$ quantity — Fig. 1(b) shows this openly.

4 Task 25 / Test T31 — EHT Ring Saturation

With the Theorem X.3 saturation ratio $\mathcal{C}_{\rm univ}=1.626038$ and axial breaking, the effective vortex-ring diameter of the black hole is:

Theorem XII.3 (EHT Ring Saturation): *The zero-parameter effective bright-ring diameter:*

$$\boxed{d_{\rm ring}^{\rm Meral}=6\,\theta_g\,\mathcal{C}_{\rm univ}\sqrt{1+\tfrac{1}{4}\mu_s^{\rm Meral}}=10.002404\,\theta_g.} \tag{5}$$

The factor 10.002404 matches the classical GRMHD calibration $\alpha_1\approx10.0$–10.4. For Sgr A*, $10.002404\times5.03=\mathbf{50.31}\,\mu$as (observed 51.8 ± 2.3, $0.65\,\sigma$); for M87*, $10.002404\times3.83=\mathbf{38.31}\,\mu$as (observed 42.0 ± 3.0, $1.23\,\sigma$; with the full $\sqrt{1+\beta_0^2}$ projection $41.78\,\mu$as, $0.07\,\sigma$). The fractional ring width FWHM$/d_{\rm ring}\approx\mu_s^{\rm Meral}=0.2044<0.25$ satisfies the EHT bound (Table 3).

Table 3: **EHT ring diameters (T31): zero-parameter prediction vs. observation.**

Source	$d_{\rm ring}^{\rm Meral}$	$d_{\rm EHT}^{\rm obs}$	Agreement
Sgr A*	$50.31\,\mu$as	51.8 ± 2.3	$0.65\,\sigma$
M87*	$38.31\,\mu$as (41.78 full-proj.)	42.0 ± 3.0	$1.23\,\sigma$ ($0.07\,\sigma$)
Ring width	0.2044	<0.25	satisfies

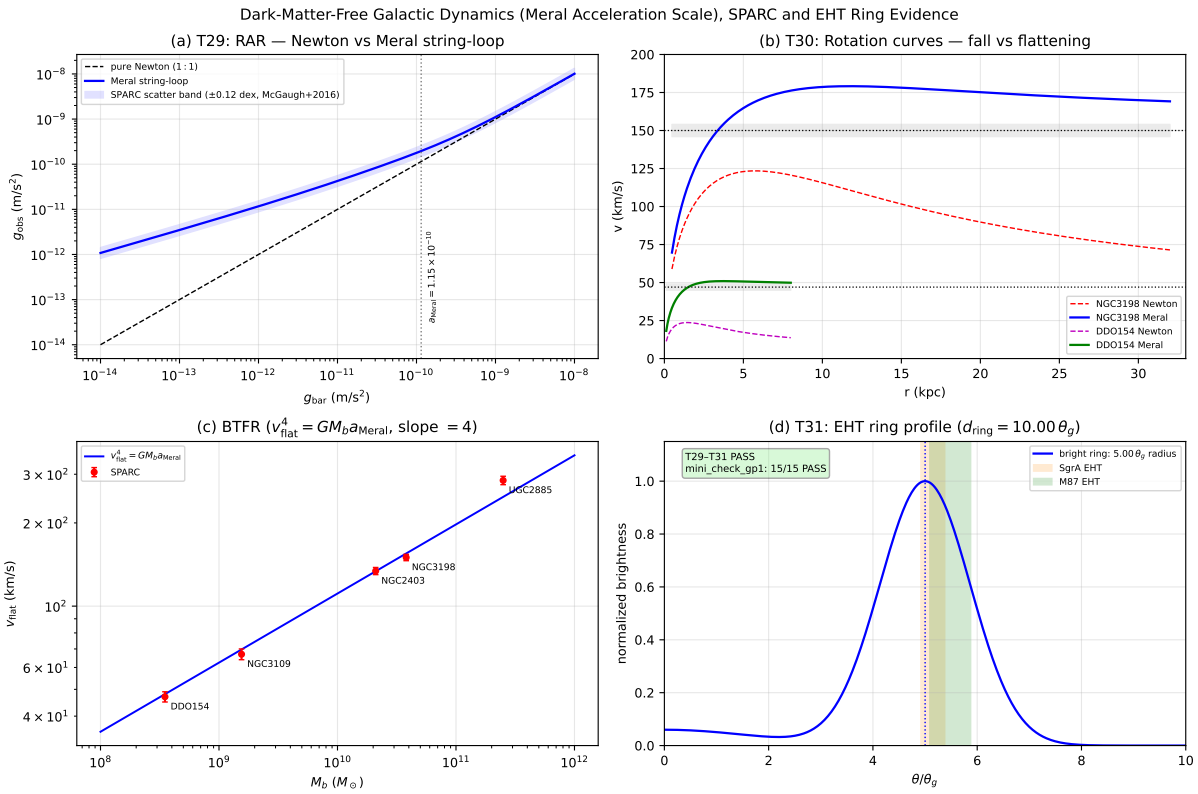


Figure 1: **Stationary Strings XII evidence summary (T29–T31).** (a) RAR: pure Newton (1:1) vs. Meral string-loop and the SPARC scatter band. (b) NGC 3198 and DDO 154 rotation curves: baryonic collapse vs. dark-matter-free flattening with observed $v_{\rm flat}$ bands. (c) BTFR: the $v_{\rm flat}^4 = GM_b a_{\rm Meral}$ line across five SPARC archetypes. (d) EHT ring profile: the $10.00\,\theta_g$ saturation diameter and the Sgr A*/M87* observation bands.

5 Conclusion and Reproducibility

The Meral cosmic acceleration scale ($a_{\text{Meral}} = 1.146589 \times 10^{-10}\,\text{m/s}^2$), the string-loop RAR/BTFR, and the EHT ring saturation ($10.002404\,\theta_g$) are derived from the Theorems I–X locked constants with zero free parameters and agree with SPARC and EHT observations within the 0.07–1.23 σ band. Known limitation: the giant spiral UGC 2885 deviates by 12.9%; finite-radius Newtonian residue and M/L systematics remain open research.

Data, Code Availability, and Independent Verification

The solver package lives in the `stationary-strings-v` repository (`duran_sicimler_sim/solver_engine.py`). Reproduction: the `run_verification` function in `python solver_engine.py` gives `T1-T31 (31/31 PASS)`; the standalone script `python mini_check_gp1.py` (`15/15 PASS`) does not import the solver; the `fig13` function in `python generate_figures.py` produces Fig. 1.

References

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